

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	SHAFFRIA.A	Reg. No.:	192571047
Section / Batch:	B. TECH CSBS	Date:	22.07.2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I SHAFFRIA.A (192571047)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge

Course Coordinator

Head of Department

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q75. Application: Multimedia Streaming Platform

A streaming platform divides media data into two equal parts and processes them recursively, with extra linear work at each stage: $T(n) = 2T(n/2) + n$. State the Master Theorem, explain the significance of a , b , and $f(n)$, and apply it to determine the time complexity. Also analyze the space complexity of this recursive streaming process.

Q76. Application: Insurance Claim Verification System

An insurance system uses three nested loops to verify policy details, claim history, and supporting records. Recall how triple nested loops affect complexity, explain the rate of growth, and determine both the time complexity and space complexity for this process.

Q75: MULTIMEDIA STREAMING PLATFORM.

A multimedia streaming platform divides large media data into two equal parts and processes them recursively, performing extra linear work at each stage. The recurrence relation is:

$$T(n) = 2T(n/2) + n$$

This represents a divide-and-conquer approach where the problem of size n is split into two subproblems of size $n/2$, and additional linear work n is done to merge or process results. This is similar to how streaming services handle encoding, buffering, or parallel data processing.

Solution Using Master Theorem:

The recurrence relation fits the form:

$$T(n) = aT(n/b) + f(n)$$

Here:

- $a = 2$ (two subproblems)
- $b = 2$ (problem size halved each time)
- $f(n) = n$ (extra linear work per level)

Now compute: $\log_b a = \log_2 2 = 1$.

Compare $f(n)$ with $n^{\log_b a}$:

- $f(n) = n$
- $n^{\log_b a} = n^1 = n$

Since $f(n) = \Theta(n^{\log_b a})$, by Case 2 of the Master Theorem:

$$T(n) = \Theta(n \log n)$$

Time Complexity:

$$T(n) = O(n \log n)$$

This means the streaming platform's recursive processing grows proportionally with $n \log n$, which is efficient for large-scale data handling.

Space Complexity:

Recursion depth = $\log_2 n$. At each level, temporary storage may require $O(n)$. Thus, overall Space Complexity = $O(n)$.

Final Answer:

The multimedia streaming platform recurrence relation $T(n) = 2T(n/2) + n$ resolves to:

- Time Complexity: $O(n \log n)$
- Space Complexity: $O(n)$

This shows the system is efficient, scalable, and well-suited for real-world streaming applications.

Q76: INSURANCE CLAIM VERIFICATION SYSTEM

An insurance claim verification system must efficiently process and validate claims submitted by policyholders. The system divides the verification process into multiple sub-tasks such as checking policy details, validating claim documents, and cross-verifying with external records. Each sub-task contributes to the overall complexity of the system. The recurrence relation is given as:

$$T(n) = T(n - 1) + O(1)$$

This indicates that each claim verification depends on the previous claim, with constant additional work at each step.

Solution:

The recurrence relation can be solved as follows:

$$T(n) = T(n - 1) + O(1)$$

Expanding step by step:

$$\begin{aligned}T(n) &= T(n - 1) + 1 \\&= T(n - 2) + 1 + 1 \\&= T(n - 3) + 1 + 1 + 1 \\&= \dots = T(1) + (n - 1)\end{aligned}$$

Thus,

$$T(n) = O(n)$$

Time Complexity:

The time complexity of the insurance claim verification system is $O(n)$. This means the system scales linearly with the number of claims, making it predictable and manageable for large datasets.

Space Complexity:

Since the recurrence relation only depends on the previous claim and requires constant additional work, the space usage is minimal. Therefore, Space Complexity = $O(1)$.

Final Answer:

The insurance claim verification system recurrence relation $T(n) = T(n - 1) + O(1)$ resolves to:

- Time Complexity: $O(n)$

- Space Complexity: $O(1)$

This demonstrates that the system is efficient, requiring linear time to process claims while using constant space, making it suitable for large-scale insurance operations.
